

in IPAbet keystrokes

CONSONANTS (PULMONIC)

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b p b		t d t d			ʈ ɖ tʀ dʀ	c ɟ c ɟ	k ɡ k ɡ	q ɢ q ɢ		ʔ ʔ
Nasal	m m	ɱ mʋ	n n			ɳ nʀ	ɲ nɟ	ŋ ŋ	ɴ nɔ		
Trill	ʙ rʙ		ʀ r						ʀ rɢ		
Tap or Flap		ⱱ ɹv	ɾ ɹ			ɽ ɽ					
Fricative	ɸ β pʰ bʱ	f v f v	θ ð tʰ dʰ	s z s z	ʃ ʒ ʃʰ ʒʰ	ʂ ʐ sʀ zʀ	ç ʝ cʰ ɟʱ	x ɣ x ɣ	χ ʁ qʰ rɔ	ħ ʕ ʕʰ ʕʱ	h ɦ h ɦ
Lateral fricative			ɬ ɮ sɭ zɭ								
Approximant		ʋ vʱ	ɹ rʱ			ɻ rʀ	j j	ɰ wʷ			
Lateral approximant			l l			ɭ lʀ	ʎ ɭɟ	ʟ lɢ			

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible

CONSONANTS (NON-PULMONIC)

Clicks

⊙	Bilabial	pC
	Dental	tC
!	(Post)alveolar	qC
‡	Palatoalveolar	cC
	Alveolar lateral	lC

Voiced implosives

b	Bilabial
d	Dental/alveolar
ʃ	Palatal
g	Velar
ŋ	Uvular

Ejectives

p' Bilabial
[p ɣ q]
t' Dental/alveolar
[t ɣ q]
k' Velar
[k ɣ q]
s' Alveolar fricative
[s ɣ q]

OTHER SYMBOLS

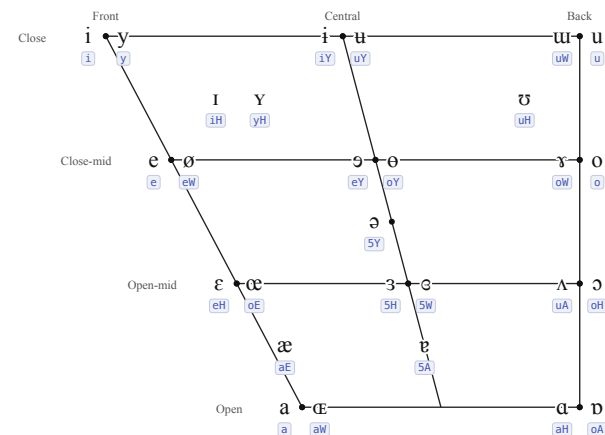
A	Voiceless labial-velar fricative	
W	Voiced labial-velar approximant	
U	Voiced labial-palatal approximant	
H	Voiceless epiglottal fricative	
Ç	Voiced epiglottal fricative	
ʔ	Epiglottal plosive	

ç	Voiceless alveolo-palatal fricative [sʝ]
ʒ	Voiced alveolo-palatal fricative [zʝ]
ɺ	Voiced alveolar lateral flap [4L]
ʈʂ	Affricate (tie bar) [tʰɕj s]
ɸ	Simultaneous f and x [xH]

DIACRITICS

☐ Voiceless [t̚k]	☐ Breathely voiced [t̚u]	☐ Dental [t̚t]
☐ Voiced [t̚v]	☐ Creaky voiced [t̚n]	☐ Apical [t̚d]
☐ Aspirated [t̚z h]	☐ Linguolabial [t̚l]	☐ Laminal [t̚ɖ]
☐ More rounded [t̚w]	☐ Labialized [t̚w]	☐ Nasalized [t̚n]
☐ Less rounded [t̚w]	☐ Palatalized [t̚z j]	☐ Nasal release [t̚z n]
☐ Advanced [t̚ɪ=]	☐ Velarized [t̚z g ɪH]	☐ Lateral release [t̚z l]
☐ Retracted [t̚ɹ=]	☐ Pharyngealized [t̚z 3 ɪH]	☐ No audible release [t̚p]
☐ Centralized [t̚u]	☐ Velarized or pharyngealized [t̚ɪy]	
☐ Mid-centralized [t̚ɤ]	☐ Raised [t̚ɪg]	
☐ Syllabic [t̚s]	☐ Lowered [t̚ɔ]	
☐ Non-syllabic [t̚o]	☐ Advanced tongue root [t̚ɪh]	
☐ Rhoticity [t̚r]	☐ Retracted tongue root [t̚h]	

VOWELS



Where symbols appear in pairs, the one to the right represents a rounded vowel.

SUPRASEGMENTALS

- Primary stress
[ˈ]
- Secondary stress
[ˌ]
- Long
[ː]
- Half-long
[ˑ]
- Extra-short
[̚]
- Minor (foot) group
[]
- Major (intonation) group
[]
- Syllable break
[.]
- Linking (absence of a break)
[əj əj]

TONES AND WORD ACCENTS

LEVEL	CONTOUR
ē [ɤːe] Extra high	ē [ɤː] Rising
ɪ [ɤ] Extra high (tone letter)	ɪ [ɤː ɤ] Rising (tone letter)
ē [ɤe] High	ē [ɤː] Falling
ɪ [ɤ] High (tone letter)	ɪ [ɤ ɤː] Falling (tone letter)
ē [ɤa] Mid	ē [ɤe ɤːe] High rising
ɪ [ɤ] Mid (tone letter)	ɪ [ɤa ɤ] High rising (tone letter)
ē [ɤː] Low	ē [ɤːː ɤː] Low rising
ɪ [ɤː] Low (tone letter)	ɪ [ɤː ɤː] Low rising (tone letter)
ē [ɤːː] Extra low	ē [ɤa ɤe ɤa] Rising-falling
ɪ [ɤː] Extra low (tone letter)	ɪ [ɤː ɤa ɤː] Rising-falling (tone letter)
ɪ [ɤː] Downstep	↗ [ɤː7] Global rise
↑ [ɤː6] Upstep	↘ [ɤː7] Global fall